Task 1: Simulating the basic quantum gates

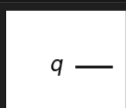
Using Qiskit and vscode, I simulated the operations of the following basic gates: Pauli X,Y, and Z, Hadaman gate, and CNOT gate, illustrating how the circuit looks and how the state vector looks on the Bloch sphere. Then, I tested the application of multiple gates on a single kit vector:

```
from qiskit import QuantumCircuit
from qiskit_aer import Aer
from qiskit.visualization import plot_bloch_multivector
from qiskit.quantum_info import Statevector
```

[3] ✓ 0.4s

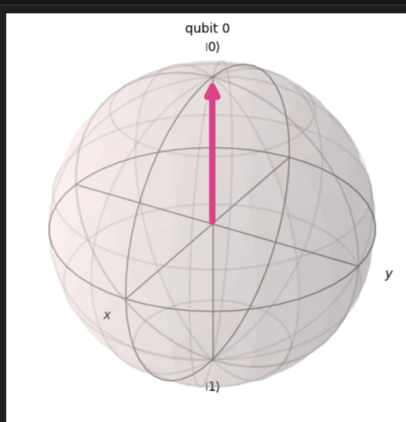
```
qc = QuantumCircuit(1)
qc.draw("mpl")
```

[4] ✓ 0.3s




```
state = Statevector.from_instruction(qc)
plot_bloch_multivector(state)
```

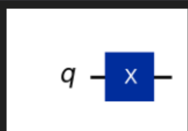
[5] ✓ 0.0s



Pauli-X gate

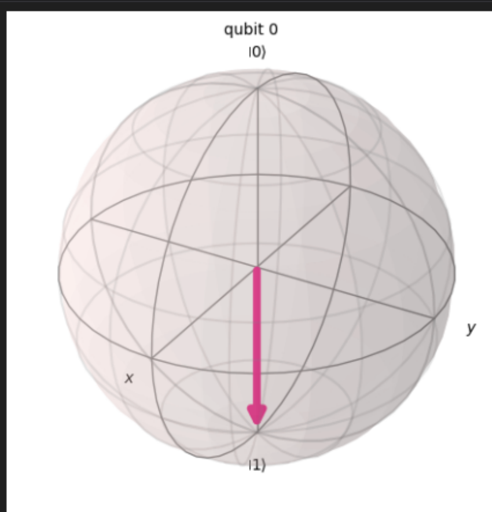
```
qc = QuantumCircuit(1)
qc.x(0)

qc.draw("mpl")
```

[6] ✓ 0.0s




```
state = Statevector.from_instruction(qc)
plot_bloch_multivector(state)
```

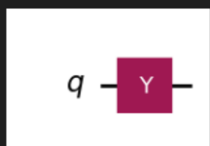
[7] ✓ 0.0s



Pauli-Y gate

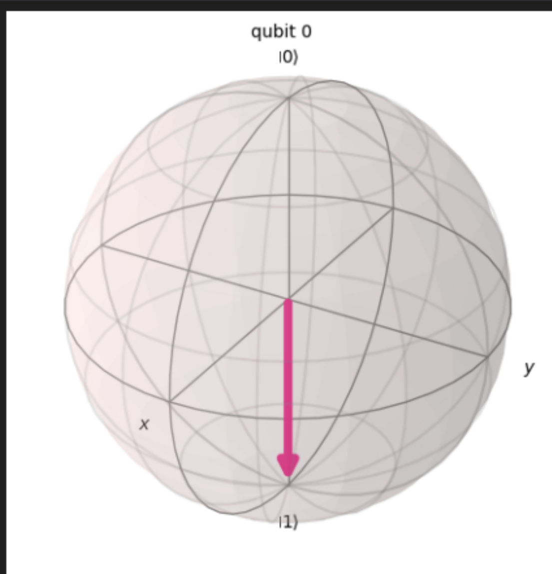
```
qc = QuantumCircuit(1)
qc.y(0)
qc.draw("mpl")
```

[8] ✓ 0.0s




```
state = Statevector.from_instruction(qc)
plot_bloch_multivector(state)
```

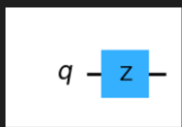
[9] ✓ 0.0s



Pauli-Z gate

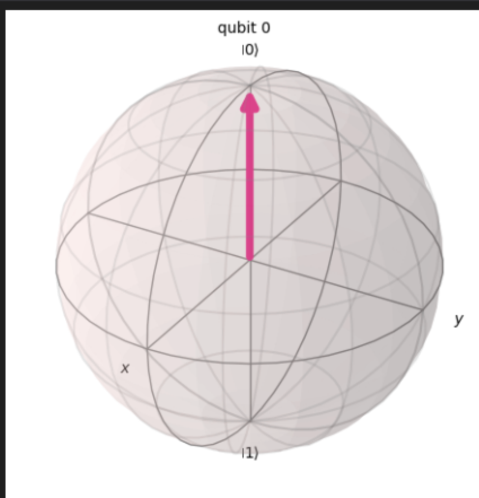
```
qc = QuantumCircuit(1)
qc.z(0)
qc.draw("mpl")
```

[10] 0.0s




```
state = Statevector.from_instruction(qc)
plot_bloch_multivector(state)
```

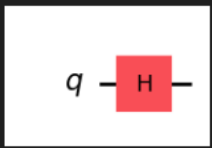
[11] 0.0s



Hadaman gate

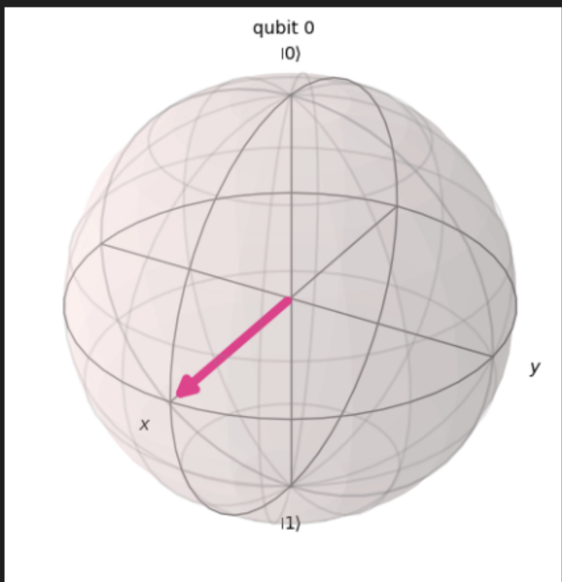
```python
qc = QuantumCircuit(1)
qc.h(0)
qc.draw("mpl")
```

[12] ✓ 0.0s




```python
state = Statevector.from_instruction(qc)
plot_bloch_multivector(state)
```

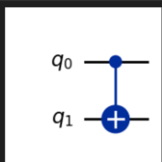
[13] ✓ 0.0s



Controlled-Not gate (CNOT)

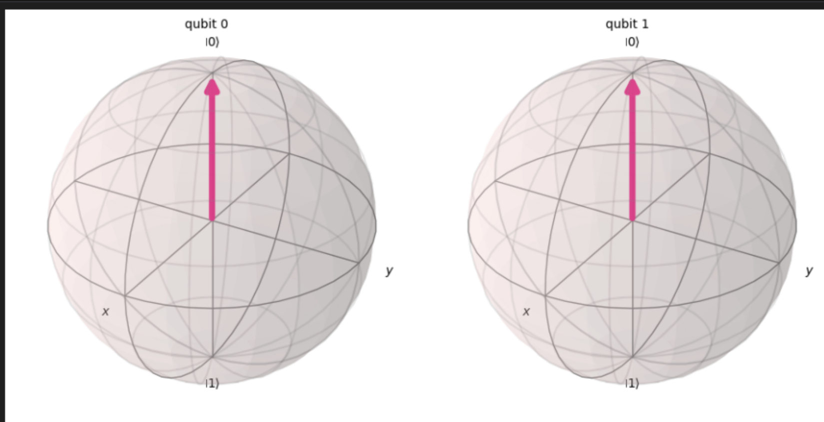
```
qc = QuantumCircuit(2)
qc.cx(0, 1)  # control = qubit 0, target = qubit 1

qc.draw("mpl")
```

[14] ✓ 0.0s




```
state = Statevector.from_instruction(qc)
plot_bloch_multivector(state)
```

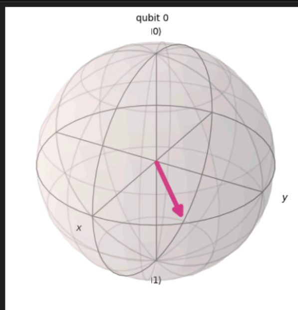
[15] ✓ 0.1s



Trying out gates with arbitrary amplitudes

```
import numpy as np
from qiskit.visualization import plot_bloch_multivector

alpha = 1/2
beta = np.sqrt(3)/2

Original_state = Statevector([alpha, beta])
plot_bloch_multivector(state)
```

43] ✓ 0.0s



amplitude pre-gates

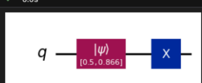
```
print(Original_state.data)
```

44] ✓ 0.0s

```
[0.5      +0.j 0.8660254+0.j]
```

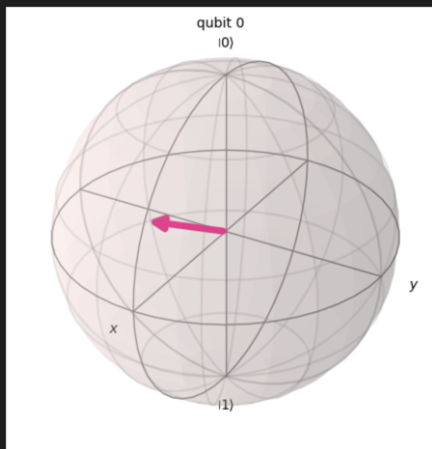
```
qc = QuantumCircuit(1)
qc.initialize(Original_state.data, 0)
qc.x(0)
qc.draw("mpl")
```

45] ✓ 0.0s



```python
state = Statevector.from_instruction(qc)
plot_bloch_multivector(state)
```

✓ 0.0s



amplitudes after applying x-gate

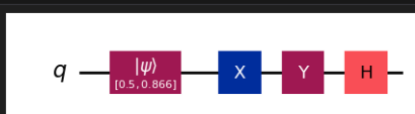
```python
print(state.data)
```

✓ 0.0s

```
[0.8660254+0.j 0.5       +0.j]
```

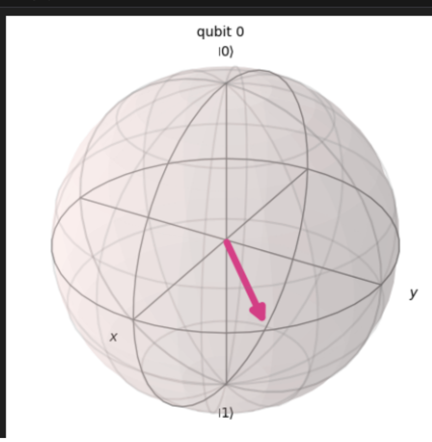
```python
qc.y(0)
qc.h(0)
qc.draw("mpl")
```

✓ 0.0s




```python
state = Statevector.from_instruction(qc)
plot_bloch_multivector(state)
```

✓ 0.1s



data after applying y and hadaman gates

```
print(state.data)
```

[50] ✓ 0.0s

```
[0.+0.25881905j 0.-0.96592583j]
```

Task 2: Quantum computing cluster plans

For the second assignment, we were task with researching available quantum computing platforms. Here are the following services I found. **NOTE:** qBraid appears to be a platform that works with other companies to provide quantum computing services, they offer direct links to said services, so one may explore them further.

IBM Quantum Platform (https://quantum.cloud.ibm.com/docs/en/guides/plans-overview)

- Has free and paid plans
- Free tier (Open plan)
  - Up to 10 minutes quantum time per 28-day rolling window.
  - Only available in the us-east region
- Paid plans:
  - Flex plan
    - Pre-purchase at least 400 minutes for jobs
      - Expire one year after purchase
    - 72$ per minute
  - Pay-As-You-Go plan:
    - Minutes are bought as they are being used (ex: if you use the service for 28 minutes, you pay only for those 28 minutes)
    - 96$ per minute
  - Premium plan
    - Enterprise level tier
    - Includes the use of Qiskit functions and Qiskit Transplier as a service
    - Minimum of 5200 minutes
    - 48$ per minutes
  - On-Prem plan

- Grants access for dedicated on-premise quantum system maintained by IBM
- “Price requires quote”?

Quantum Computing User Program Access (QCUP)

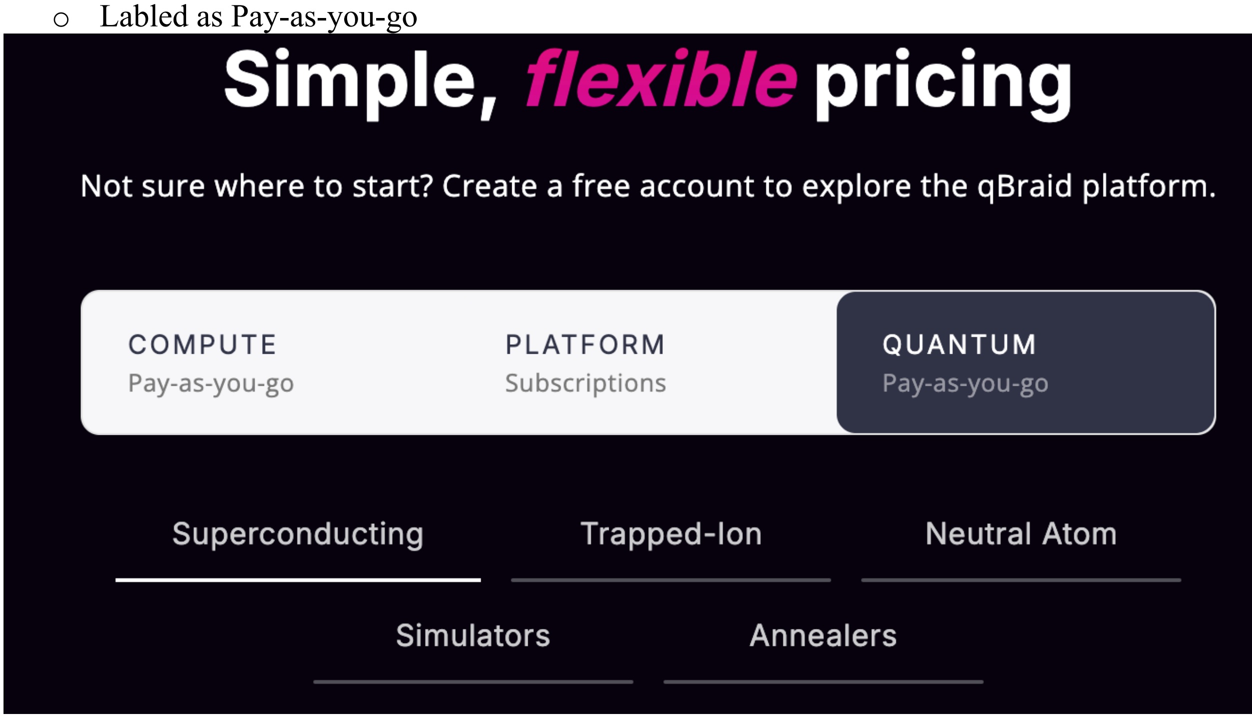
- From Oak Ridge National Laboratory
- Users must send in a proposal describing the nature and merits of the project
- The program gives access to IBM services and resource reservations
- Link to program page with steps to apply
- Unsure as to what resources they offer

Quantinuum (https://www.quantinuum.com/)

- Offers quantum computing resources but requires the filling of a forum/comunicating with their team to access, leaving this in for reference

qBraid (https://www.qbraid.com/)

- Offers many options, some of these are the quantum specific options
- Seems to be a middle man for multiple companies. Luckly, they offer direct links to the company’s site where we can see the listed services
    - Labled as Pay-as-you-go

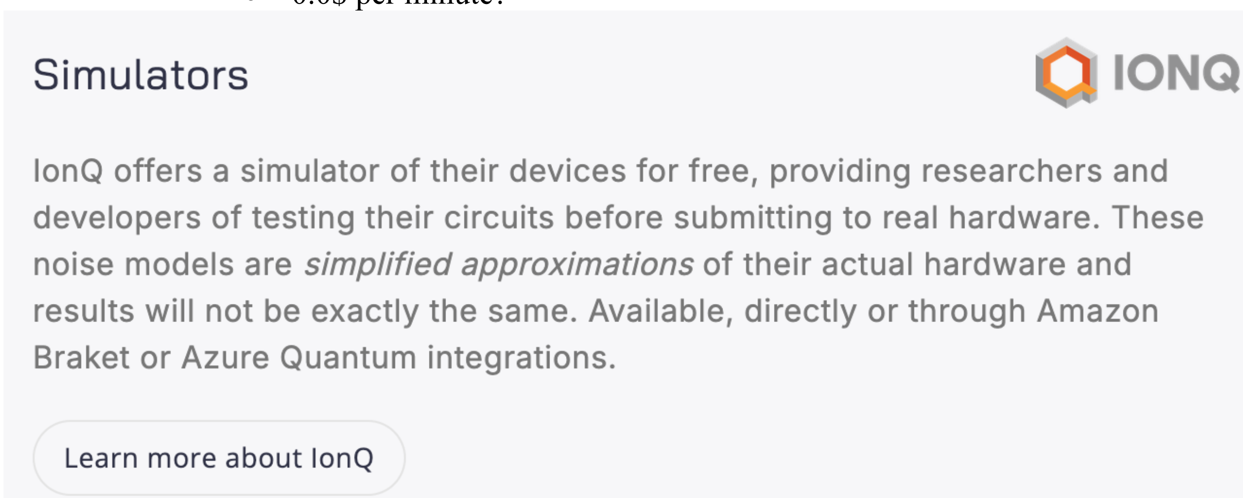


-
- Superconducting quantum computers (rigetti)

    - Available with qBraid's Amazon Braket integration and Azure Quantum integration, Rigetti's quantum processors features enhanced readout capabilities that contribute to better overall circuit fidelities.
    - Ankaa-3:
        - 0.3$ per task + 0.0009$ per shot
        - 84 qubits
        - Average T1 and T2 lifetime of 13.012 micro seconds and 11.999 microseconds respectivly
- IQM
    - Available with qBraid's Amazon Braket integration, IQM Garnet is a high-fidelity 20-qubit QPU based on superconducting transmon qubits.
    - Garnet
        - 0.3$ per task + 0.00145 per shot
        - 20 quibits
        - Average T1 and T2 lifetime of 13.012 micro seconds and 11.999 microseconds respectivly
        - Average CZ fidelity of 98.703%
- Trapped-Ion Quantum computers (ION Q)
    - Available directly on qBraid, through qBraid's Amazon Braket integration, as well as qBraid's Azure Quantum Integration, IonQ's quantum processors features trapped-ion quantum computers with all-to-all connectivity.
    - Aria 1
        - 25 quibits
        - 0.3$ per task + 0.03$ per shot
        - Average T1 and T2 lifetime 100 seconds and 1 second respectivly
        - Average Fidelity: 99.98% Single-qubit gates, 98.46% Two-qubit gates
    - Aria 2
        - 25 quibits
        - 0.3$ per task + 0.03$ per shot
        - Average T1 and T2 lifetime 10 seconds and 1.5 second respectivly
        - Average Fidelity: 99.95% Single-qubit gates, 97.35% Two-qubit gates
    - Forte-1
        - 36 quibits
        - 0.3$ per task + 0.08$ per shot
        - Average T1 and T2 lifetime 100 seconds and 1 second respectivly
        - Average Fidelity: 99.98% Single-qubit gates, 99.56% Two-qubit gates
- Trapped-Ion Quantum computers (Quantinuum)
    - Available with qBraid's Azure Quantum Integration by request, Quantinuum's quantum processors features trapped-ion quantum computers with all-to-all connectivity and mid-circuit measurement.
    - System Model H1
        - 20 qubits

        - Pricing available upon request
        - Native gate set: single-qubit rotations, two-qubit zz gates, arbitary-angle zz gates, general SU(4) entangle
        - Average Fidelity: 99.998% single-qubit gates, 99.91% two qubit gates
    - System Model H2
        - 56 qubits
        - Pricing available upon request
        - Native gate set: single-qubit rotations, two-qubit zz gates, arbitary-angle zz gates
        - Average Fidelity: 99.997% single-qubit gates, 99.87% two qubit gates
- Neutral Atom Quantum Computer (|QuEra>)
    - Available with qBraid's Amazon Braket integration, QuEra's quantum computers are based on programmable arrays of neutral atoms which use Rydberg states to create entanglement.
    - Aquila
        - 256 qubits
        - 0.3$ per task + 0.01$ per shot
        - Site position error of 0.100 micrometers
        - Atom position error of 0.200 micro meters
        - Typical atom loss probability: 0.005
- Neutral Atom Quantum Computer (PASQAL)
    - Available with qBraid's Azure Quantum integration, Pasqal's quantum computers are based on programmable arrays of neutral atoms configred in 1D or 2D lattices.
    - Fresnel
        - 100 quibits
        - Pricing available upon request
- Simulators
    - Quantum computing simulators are software programs that emulate the behavior of quantum computers on classical computing systems.qBraid
        - 0.005$ per task + 0.075$ per minute
        - Based on LLVM architecture
        - Written in Rust
    - Aws braket service:
    - Amazon Web Services (AWS) offers several quantum computing simulators through its Amazon Braket service, providing researchers and developers with powerful tools to explore and test quantum algorithms without the need for physical quantum hardware.
        - SV1
            - 0.075$ per minute
            - Max runtime of 6 hours
            - 34 qubits

            - Can run 100 concurrant tasks
        - TN1
            - 0.275$ per minute
            - Max runtime of 6 hours
            - 50 qubits
            - Can run 10 concurrant tasks
        - DM1
            - 0.075$ per minute
            - Max runtime of 6 hours
            - 17 qubits
            - Can run 50 concurrant tasks
    - IONQ
        - IONQ simulator
            - 0.0$ per minute?

- 

Simulators

IonQ offers a simulator of their devices for free, providing researchers and developers of testing their circuits before submitting to real hardware. These noise models are *simplified approximations* of their actual hardware and results will not be exactly the same. Available, directly or through Amazon Braket or Azure Quantum integrations.

Learn more about IonQ

            - Can specify between aria-1, aria-2, harmony or ideal hardware simulation
            - Up to 29 qubits
            - Noise model based on characterization
    - Quantinuum
        - H1 Syntax Checker/H2 Syntax Checker
            - If the code compiles, the syntax checker returns a success status and a result of all 0s. If the code does not compile, the syntax checker returns a failed status and give the error returned to help users debug their circuit syntax.
        - H1 emulator
            - 20 quibits
            - Native gate set: single-qubit rotation, two-qubit ZZ gates, arbitrary-angle ZZ gates, general SU(4) entangle

            - Average Fidelity: 99.998% single qubit gates, 99,91% two-qubit gates
        - H2 emulator
            - 56 quibits
            - Native gate set: single-qubit rotation, two-qubit ZZ gates, arbitrary-angle ZZ gates, general SU(4) entangle
            - Average Fidelity: 99.997% single qubit gates, 99,87% two-qubit gates
    - Neutral Atom Quantum Emulator (PASQAL)
        - Emu-TN
            - Tensor network simulator which can simulate 100 qubits
            - Runs on a cluster of A100s
- Simulated Annealers
    - NEC Vector Annealing is a quantum-inspired computing method that uses simulated annealing to solve complex optimization problems.
    - Vecotr annealer (hosted in japan)
        - Price available upon request
        - Runs on SX-Aurora TSUBASA vector supercomputer
        - 300,000 bit-scale